GINHAWA · AIR-QUALITY STATION

Station Build Manual: Diagrams

All the figures from the build manual, one per page. Follow the full manual for the steps and checks.

1. Block diagram of the station power and data paths (Stage 1: What you are building)
2. Parallel, series and short-circuit connections compared (Stage 2: Electricity in plain words)
3. Sketches of the tools needed for the build (Stage 4: Tools and workspace)
4. How to use a multimeter for voltage and continuity (Stage 6: Skill: using the multimeter)
5. Resistor colour bands for 4.7 kilo-ohm (Stage 6: Skill: using the multimeter)
6. Good and bad solder joints (Stage 7: Skill: soldering)
7. Four steps of through-hole soldering (Stage 7: Skill: soldering)
8. Four steps to solder a wire to a pad (Stage 7: Skill: soldering)
9. Practice perfboard with five soldering exercises (Stage 7: Skill: soldering)
10. Steps to strip, twist and tin a wire (Stage 8: Skill: wires, heat-shrink, headers and connectors)
11. Three steps to use heat-shrink tubing (Stage 8: Skill: wires, heat-shrink, headers and connectors)
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29. Enclosure drilling plan and internal layout (Stage 25: Prepare the enclosure)
30. Steps to drill holes and cut a window in the box (Stage 25: Prepare the enclosure)
31. Pole mounting with tilted solar panel above the station box (Stage 27: Mount it outside)

STAGE 1: WHAT YOU ARE BUILDING

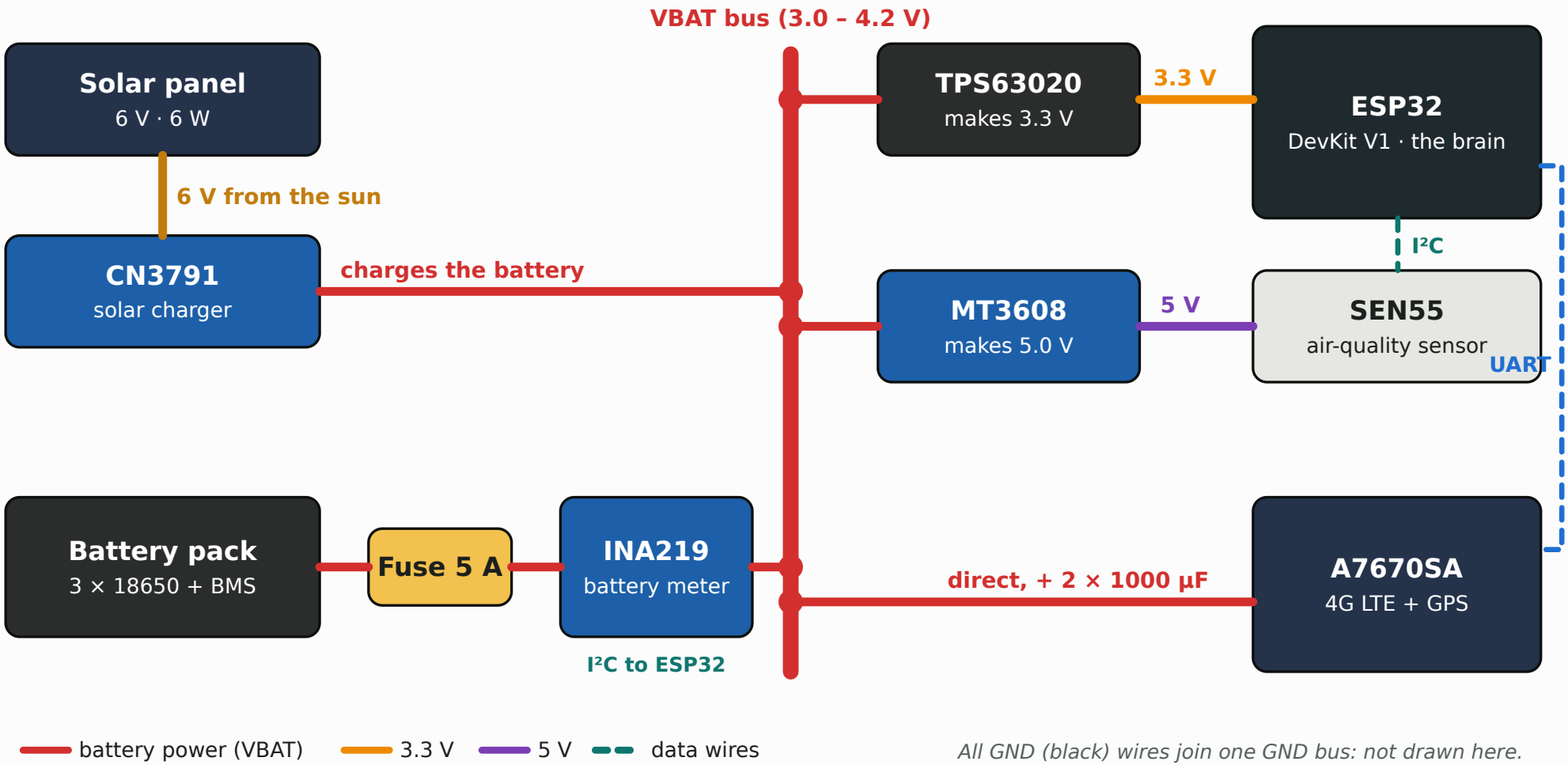
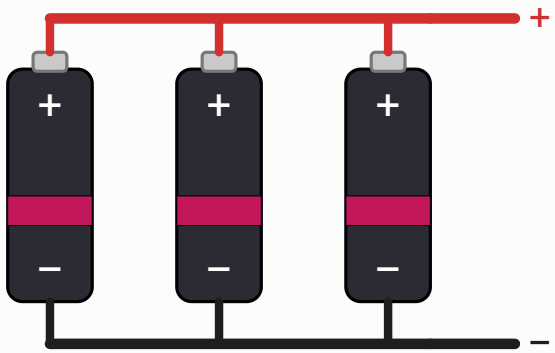


Figure 1. How power and data move through the station. The battery feeds one **VBAT bus**; every module takes its power from that bus. The solar panel refills the battery through the CN3791.

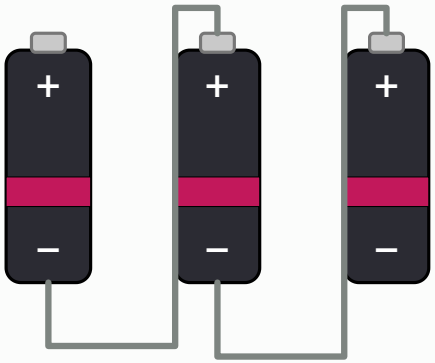
STAGE 2: ELECTRICITY IN PLAIN WORDS

Parallel: what this station uses



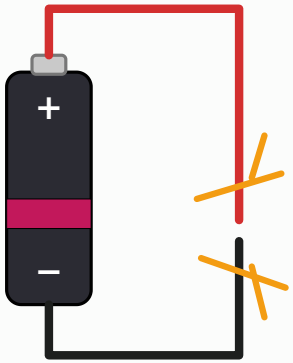
All + together, all − together.
Voltage stays 3.7 V (3.0 – 4.2 V).
Capacity adds: 3 × 2600 = 7800 mAh.

Series: NOT used here



+ of one cell to − of the next.
Voltage adds: about 11 V.
Would destroy the 3.7 V parts.

Short circuit: never



+ touches − with nothing between.
Huge current: wires melt, cell burns.
The BMS and the fuse are there to stop it.

Figure 2. Three ways cells can be connected. The station uses **parallel**: same voltage, more capacity.

STAGE 4: TOOLS AND WORKSPACE

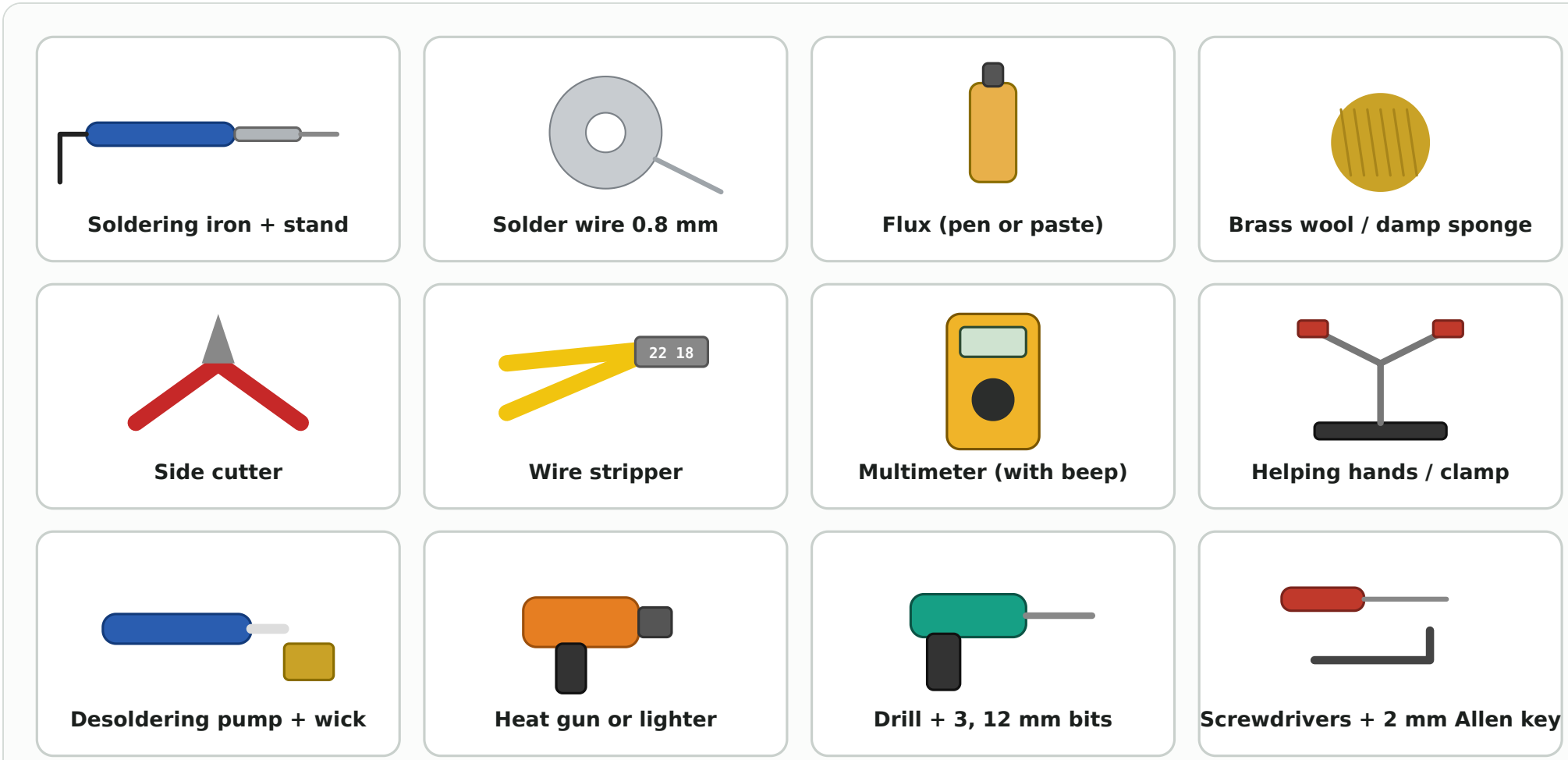


Figure 3. The tools you need. Borrow what you can; the table below says which ones you must have.

STAGE 6: SKILL: USING THE MULTIMETER

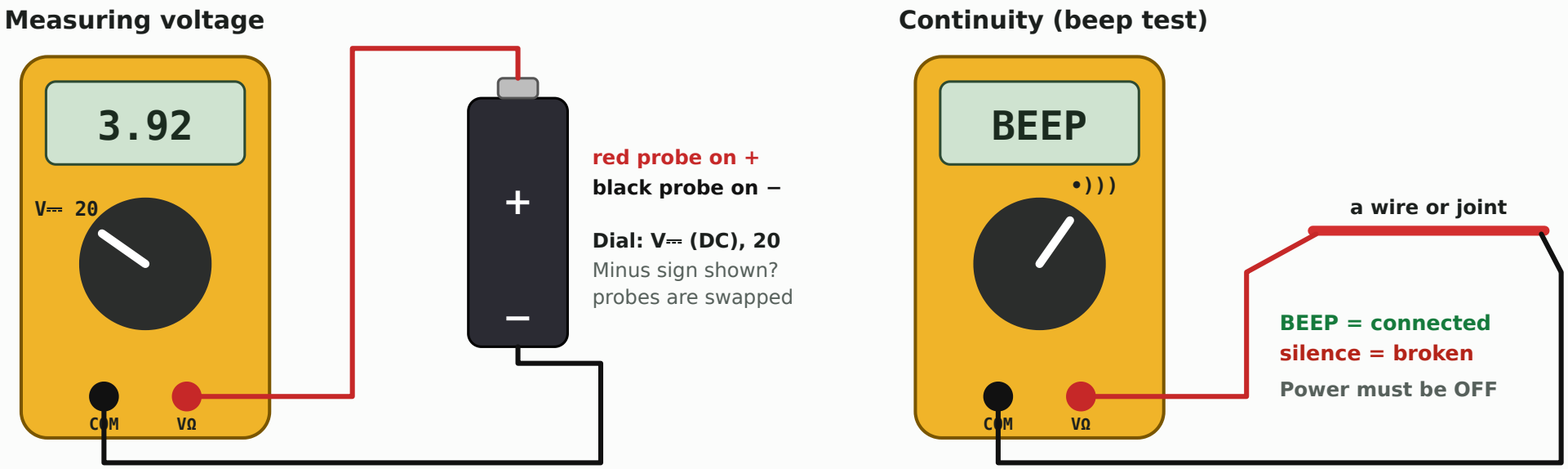


Figure 4. The two multimeter settings you need. Black probe always goes in **COM**, red in **VΩ**.

STAGE 6: SKILL: USING THE MULTIMETER

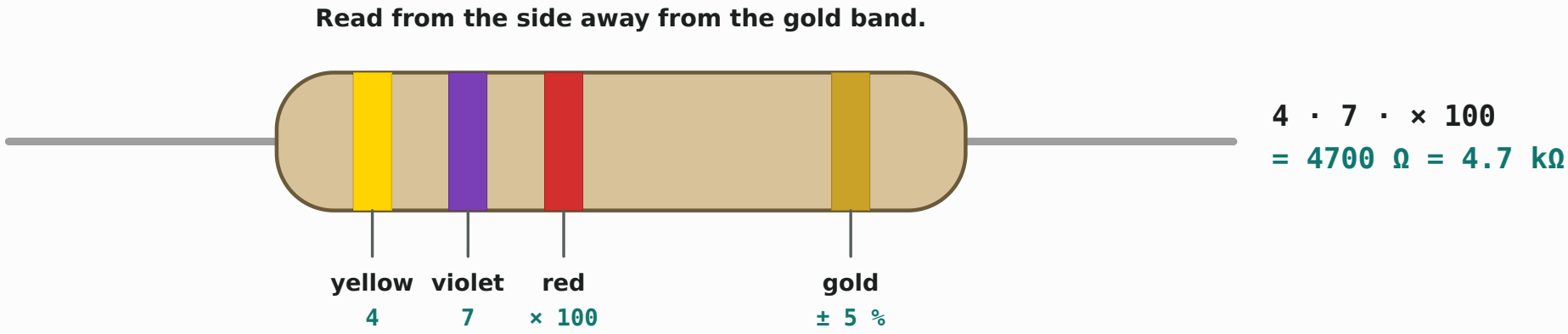


Figure 5. How to read the 4.7 kΩ resistor: **yellow, violet, red, gold**. Or measure it: the multimeter on 20k reads about 4.6 – 4.8.

STAGE 7: SKILL: SOLDERING

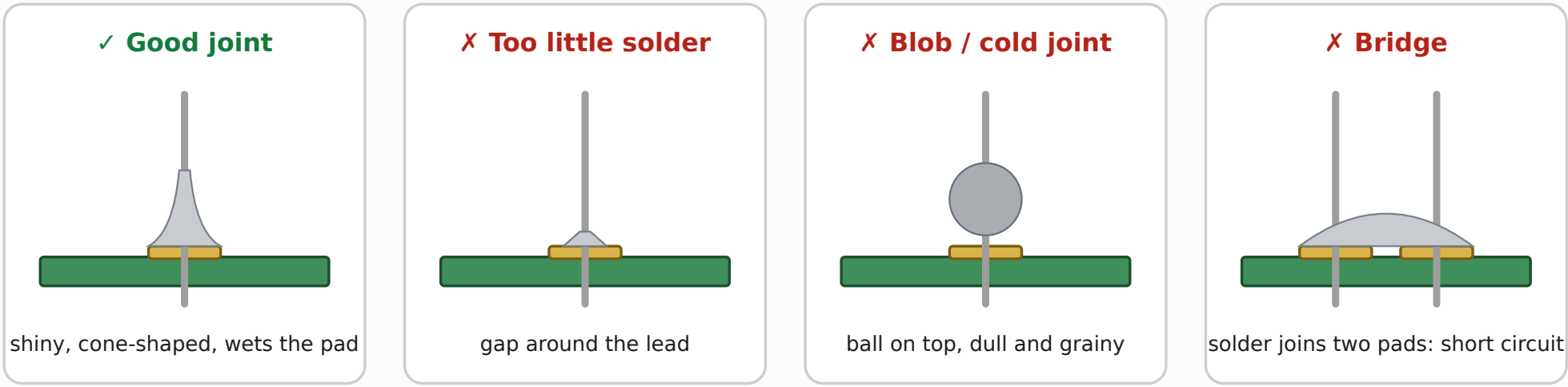


Figure 6. What solder joints should look like. Fix any ✗ joint: reheat it and add or remove solder (use the tip to drag extra away).

STAGE 7: SKILL: SOLDERING

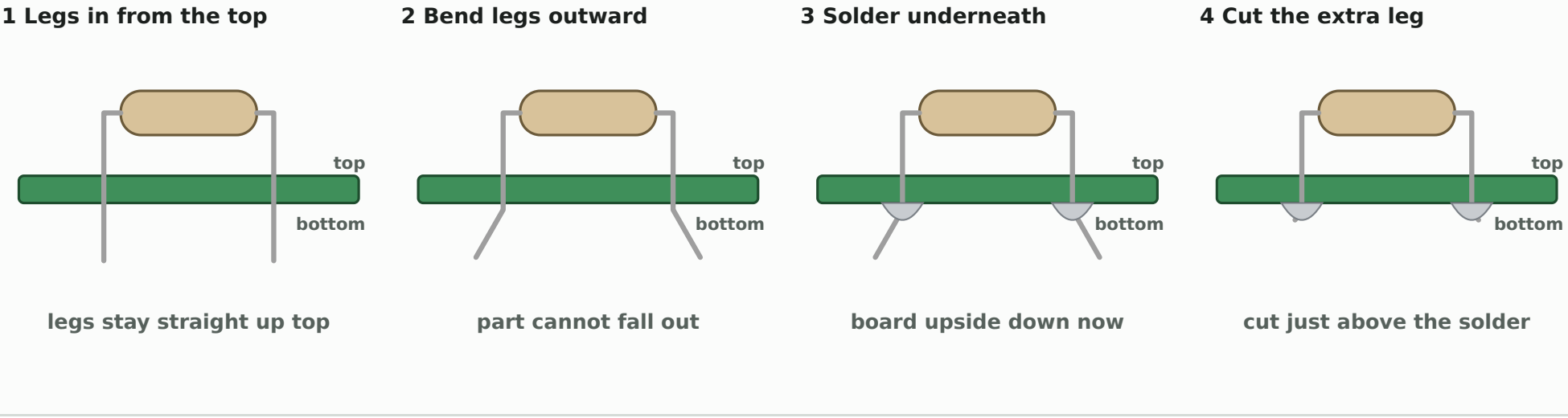


Figure 7. Soldering a part with legs (resistors, capacitors, headers) through the perfboard holes.

STAGE 7: SKILL: SOLDERING

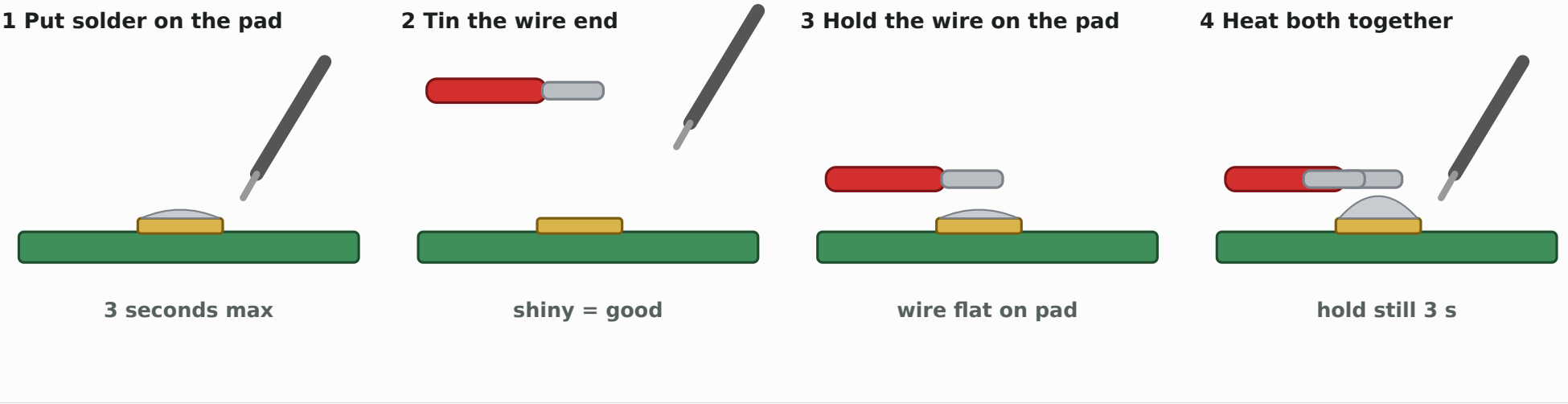
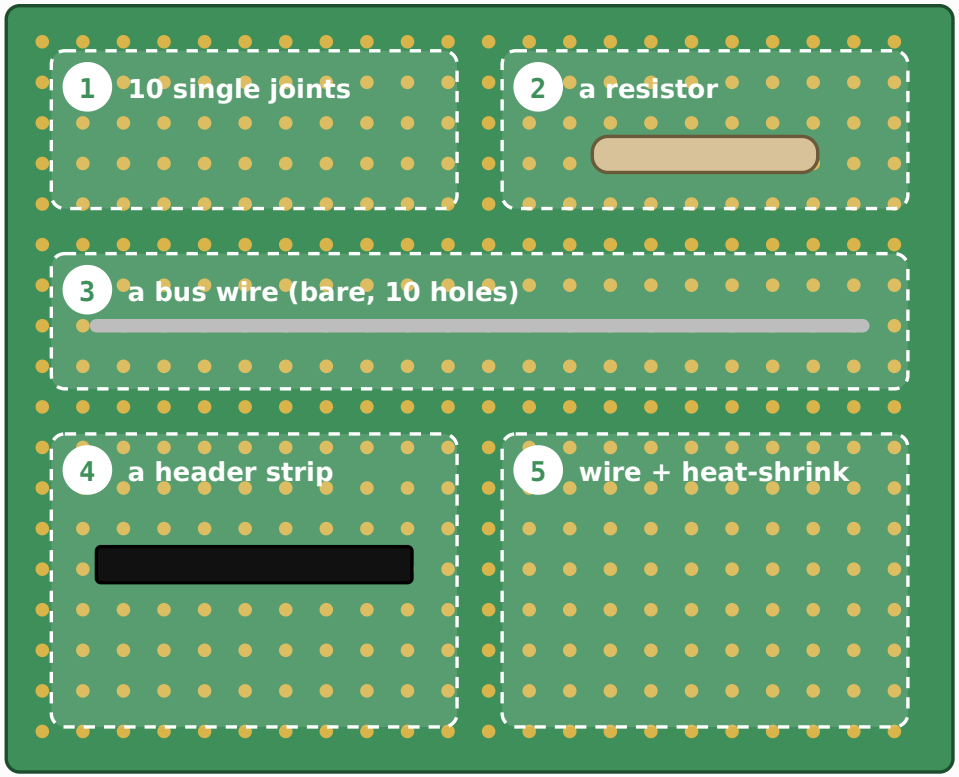


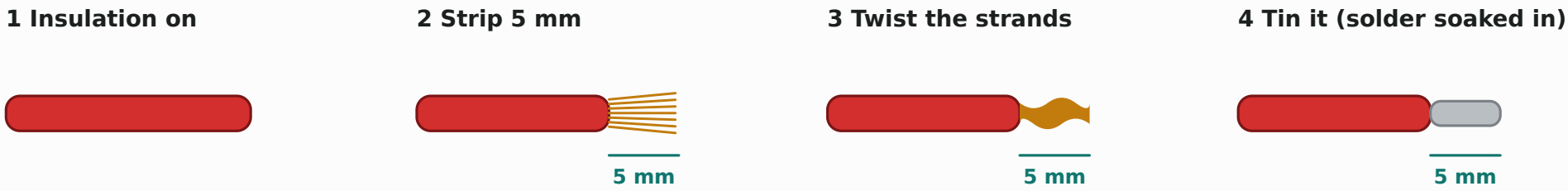
Figure 8. Soldering a wire onto a flat pad (module pads, BMS strips, bus wires). Tinning both parts first makes it quick and strong.

STAGE 7: SKILL: SOLDERING



Practice board
Use the spare (2nd) 7 × 9 cm board.
Do NOT use the 5 × 7 cm board: it becomes the power board. Practise until 9 of 10 joints look like the "good joint" drawing. This takes 30 – 60 minutes and saves the real parts.

Figure 9. Practice board: five exercises before you touch the real parts.



Wire stripper holes: put the wire in the hole with the SAME number as the wire (AWG).

18 AWG (power wire) → hole marked 18 or 1.0 mm. 22 AWG (signal wire) → hole marked 22 or 0.6 mm.

Hole too small cuts strands (wire gets weak). Hole too big only scrapes the insulation. Count the strands: none should be cut.

Figure 10. Preparing every wire end. A tinned end looks silver and stiff and never frays.

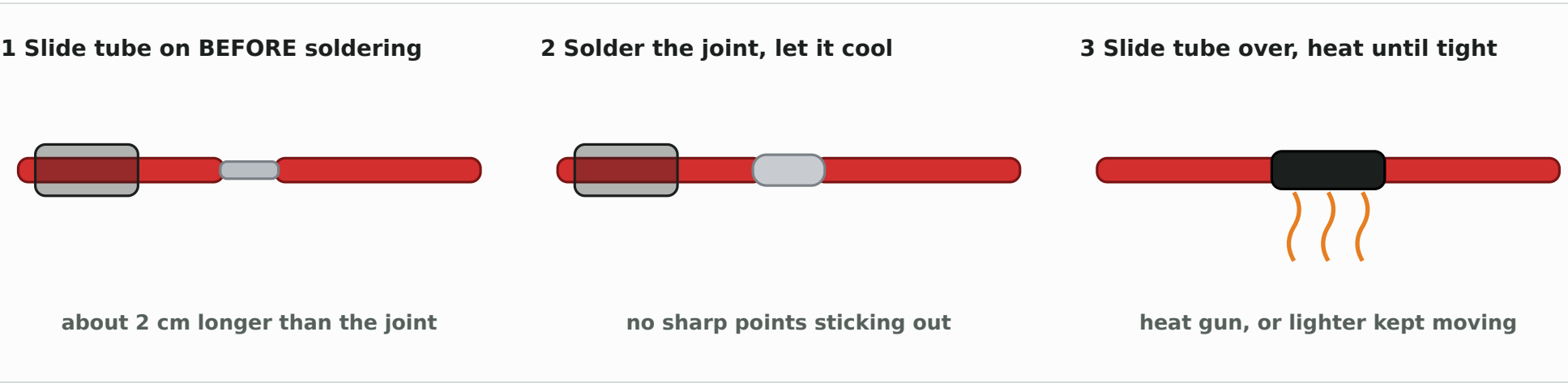


Figure 11. Heat-shrink tubing insulates a joint. It shrinks to about half its size when heated.

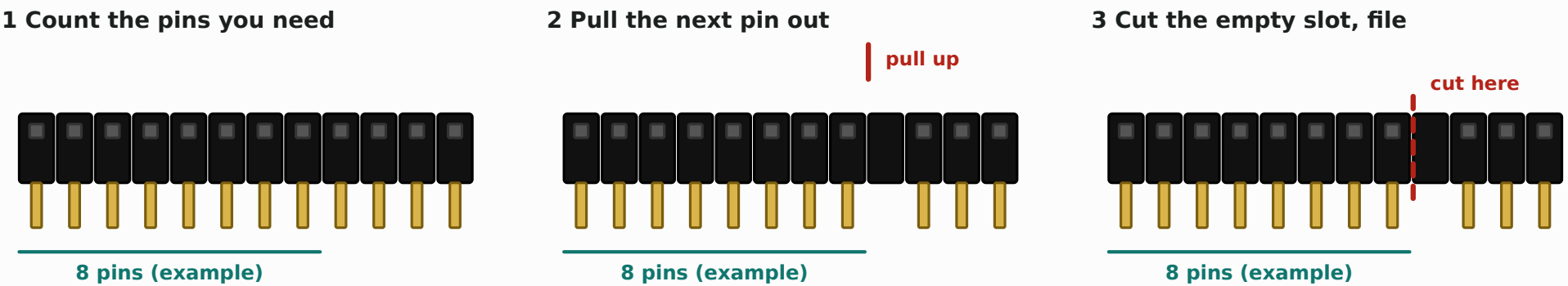
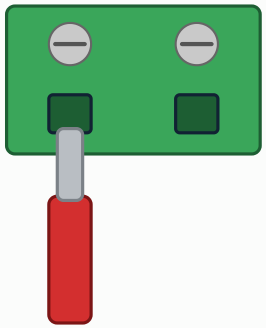


Figure 12. Cutting female headers to length. You lose one pin at each cut, so pull that pin out first: the plastic then cuts cleanly.

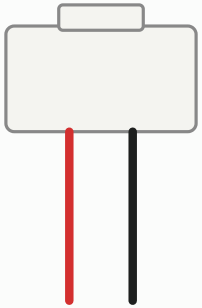
STAGE 8: SKILL: WIRES, HEAT-SHRINK, HEADERS AND CONNECTORS

Screw terminal (INA219)



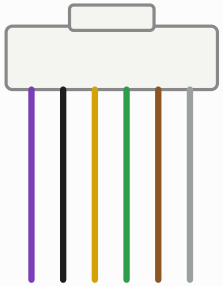
Strip 6 mm, tin, push in, tighten, then tug: must not move.

Plug cable (CN3791)



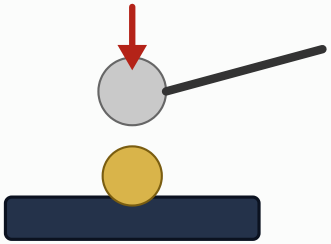
Latch side up. Push until it clicks. Pull by the plug, never the wires.

JST GH 6-pin (SEN55)



Tiny and fragile. Line it up, push straight in gently. No force.

Antenna plug (u.FL / IPEX)



Press straight down until it snaps. Lift straight up.

Figure 13. The four kinds of connectors in the station and how to handle each one.

STAGE 11: BUILD THE BATTERY PACK

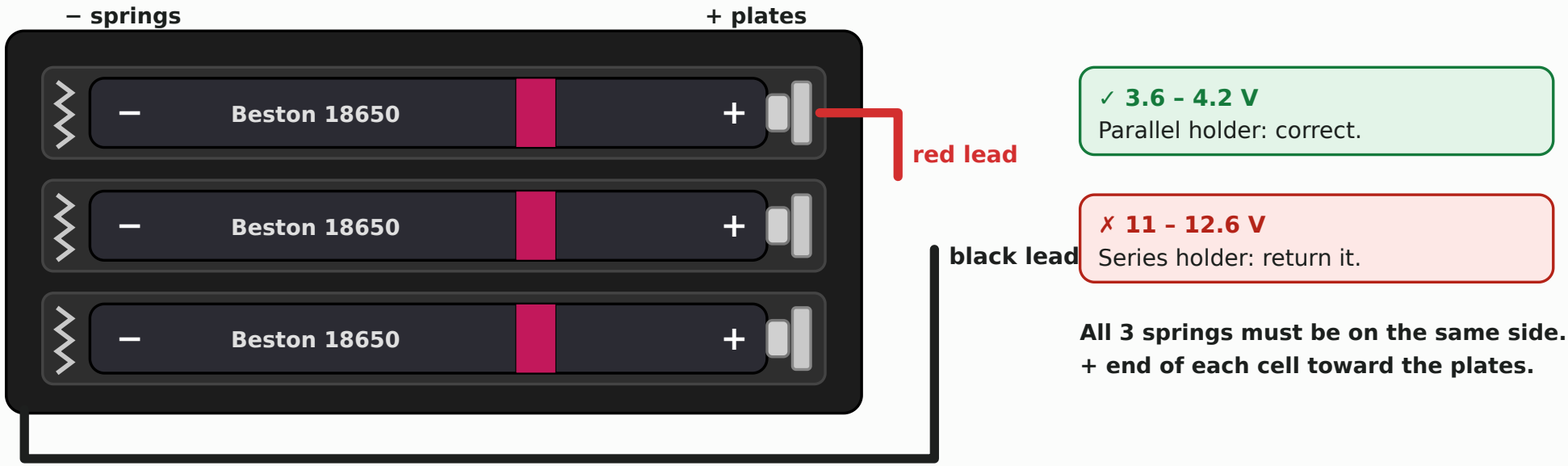


Figure 14. Cells in the 3-cell **parallel** holder. Every cell faces the same way: + end on the metal plates, − end on the springs.

STAGE 11: BUILD THE BATTERY PACK

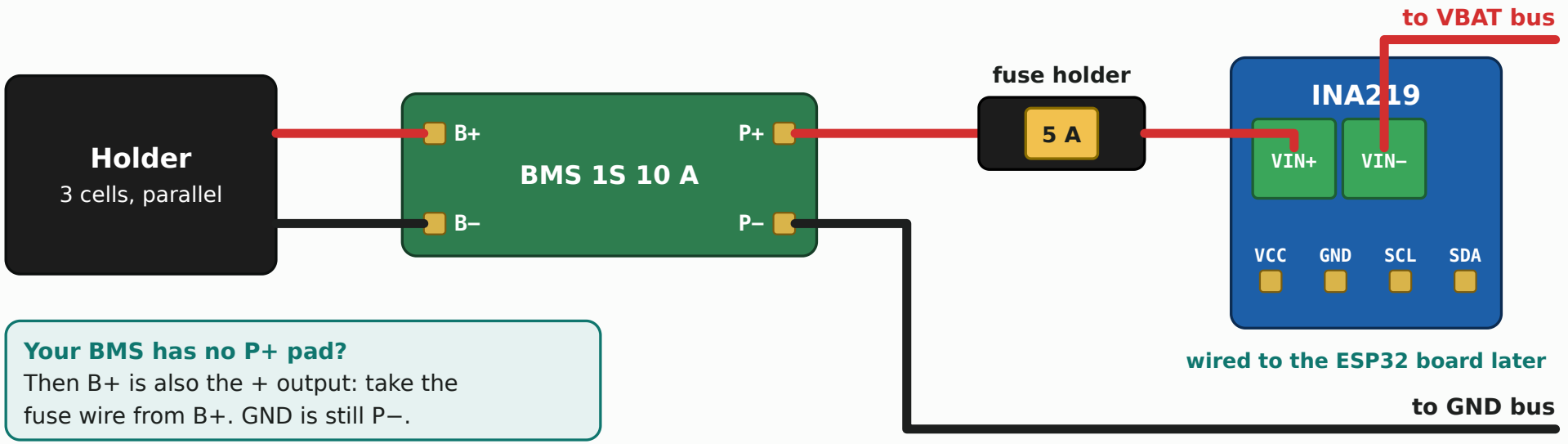


Figure 15. Battery → BMS → fuse → INA219. The thick red path carries all the station's current. **P−** becomes the station's ground (GND).

STAGE 12: MAKE THE POWER BOARD

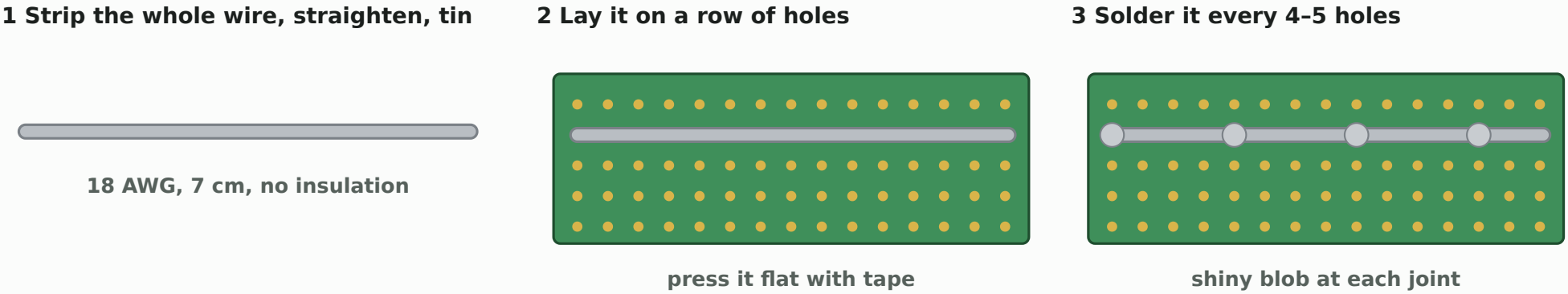
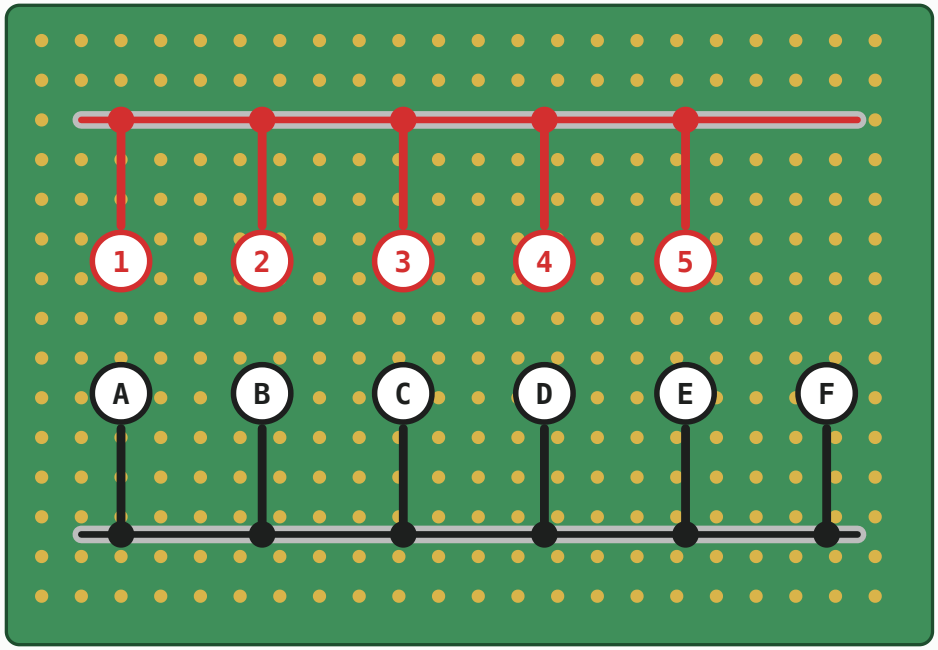


Figure 16. Making a bus wire: one long bare wire that many other wires solder onto.

STAGE 12: MAKE THE POWER BOARD

Red bus (VBAT)
1 INA219 VIN− · 18 AWG
2 CN3791 BAT +
3 TPS63020 VIN
4 MT3608 VIN+
5 A7670SA VBAT · 18 AWG



Black bus (GND)
A BMS P− · 18 AWG
B CN3791 BAT −
C TPS63020 GND
D MT3608 VIN−
E A7670SA GND · 18 AWG
F ESP32 board · 18 AWG

Power board (5 × 7 cm perfboard), seen from the top

Red bus = VBAT, black bus = GND: bare, tinned 18 AWG wire. Taps at least 3 holes apart.

Figure 17. The power board is two long bus wires. Everything that needs battery power solders to the red bus (taps 1–5); every ground solders to the black bus (taps A–F).

STAGE 14: SOLAR PANEL AND CHARGER (CN3791)

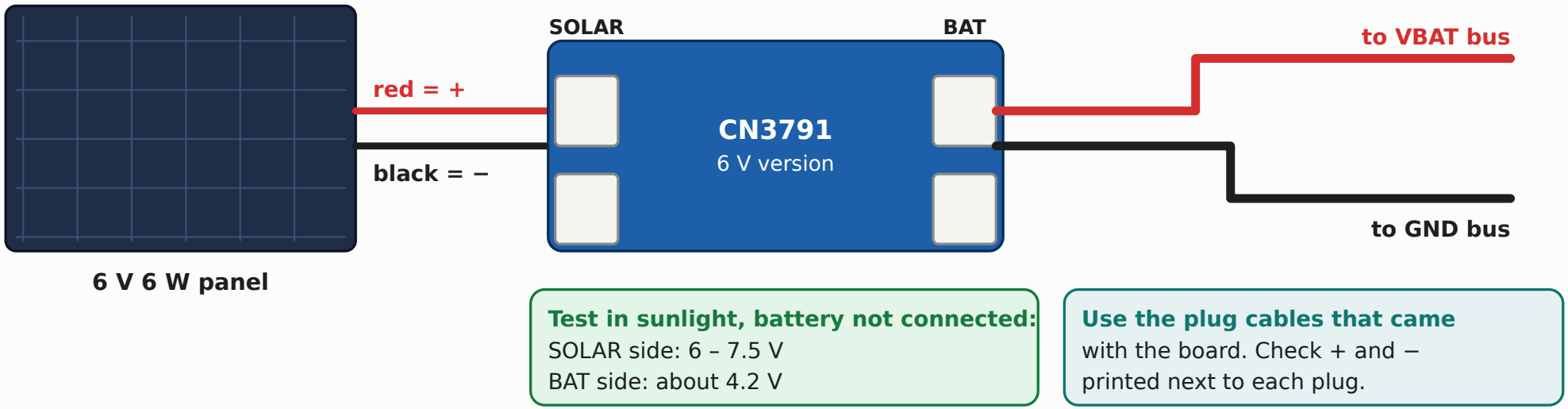


Figure 18. Solar panel → CN3791 → battery bus. Either SOLAR plug works (they are connected together), and the same for BAT.

STAGE 15: 3.3 V CONVERTER (TPS63020)

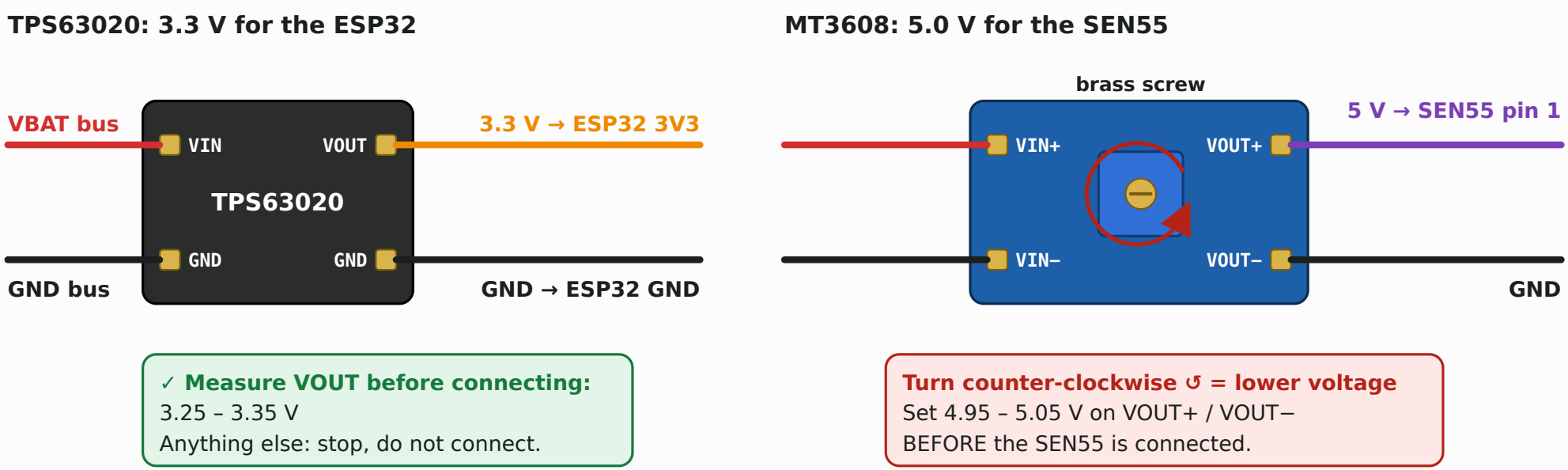


Figure 19. The two voltage converters. Both take power from the VBAT bus. Measure their outputs before anything is connected to them.

STAGE 17: MAKE THE ESP32 BOARD

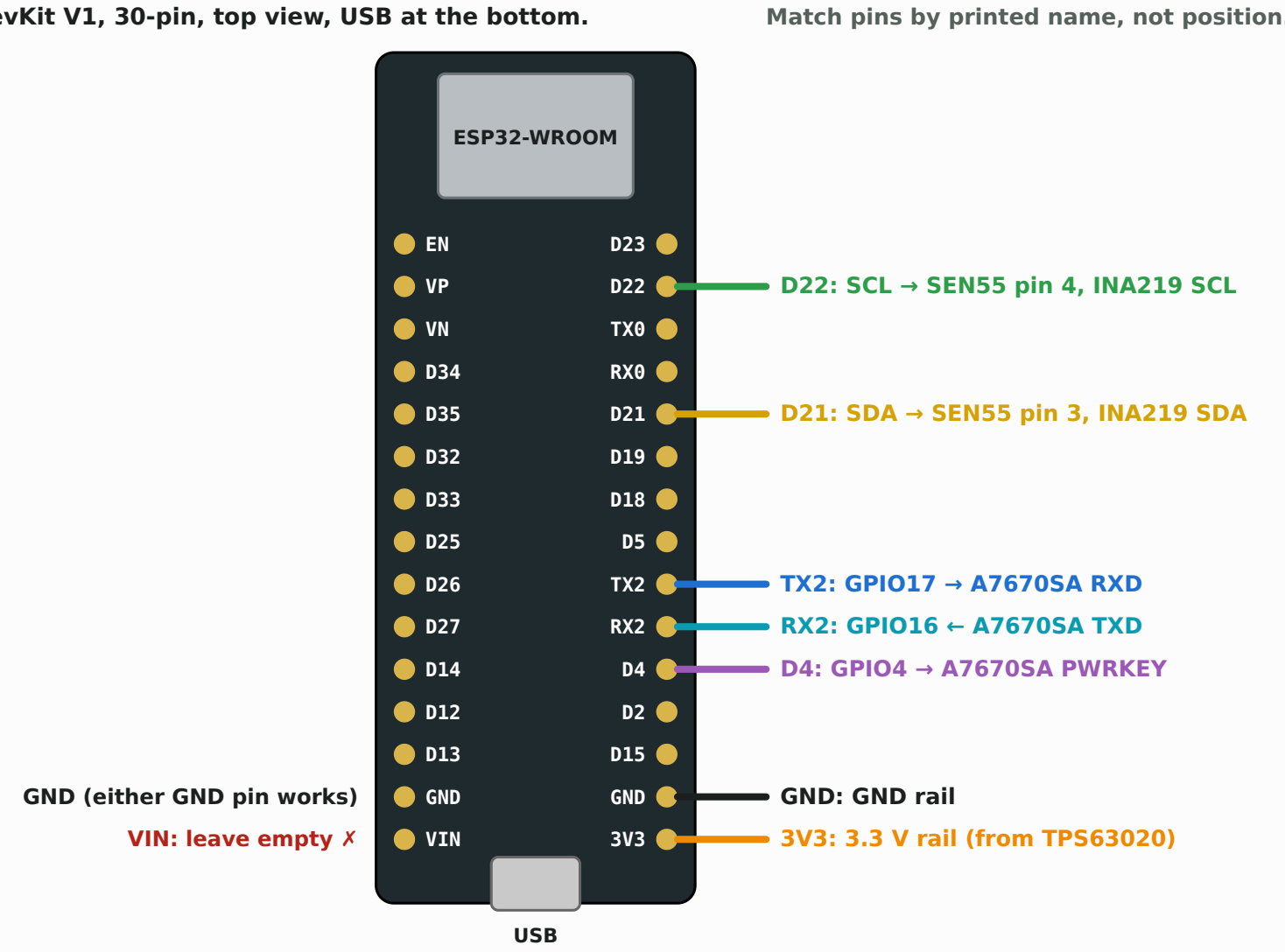


Figure 20. The ESP32 pins this project uses, all on one side of a 30-pin DevKit V1. **TX2 = GPIO17** and **RX2 = GPIO16** on the board's labels.

STAGE 17: MAKE THE ESP32 BOARD

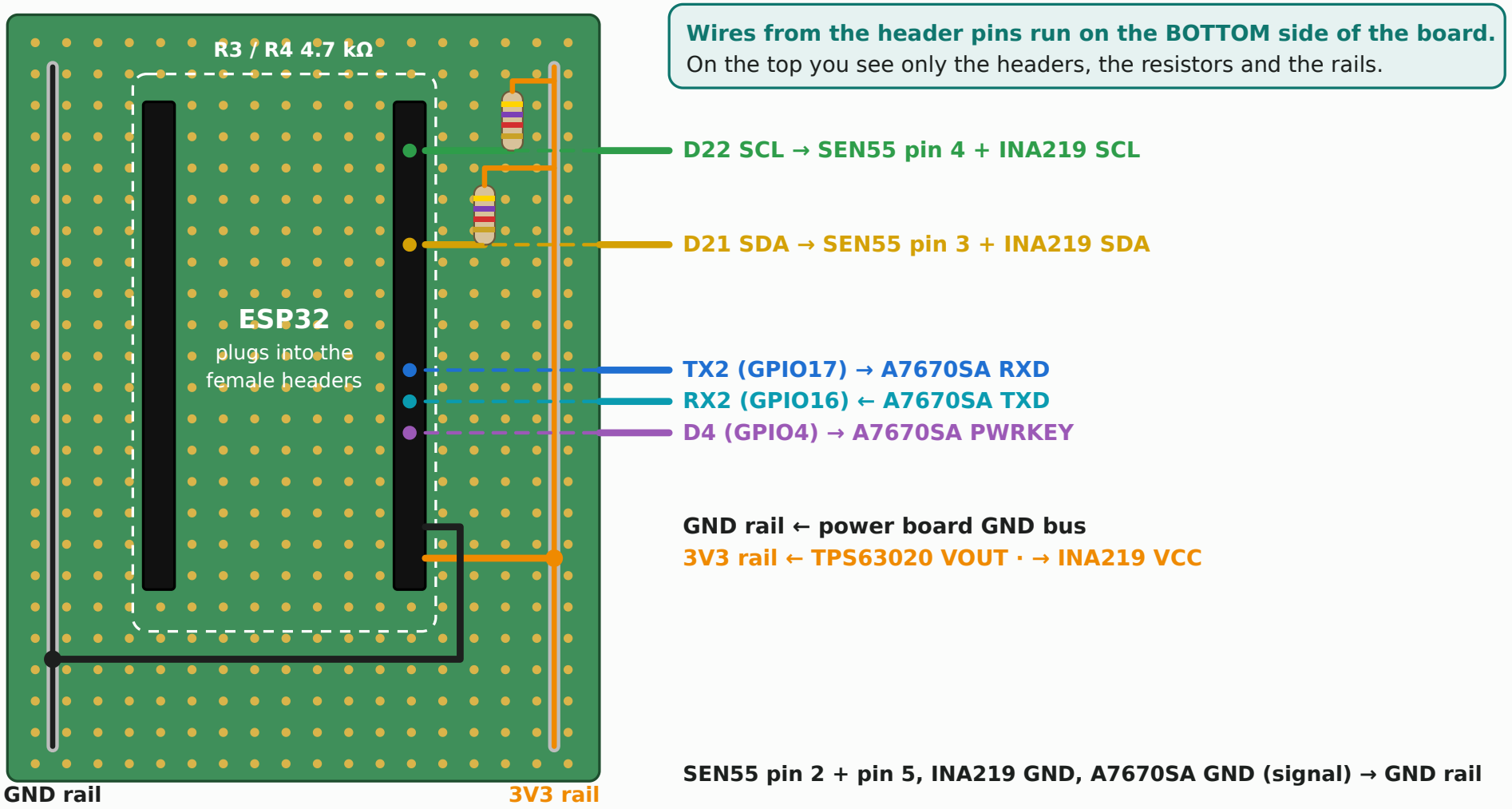
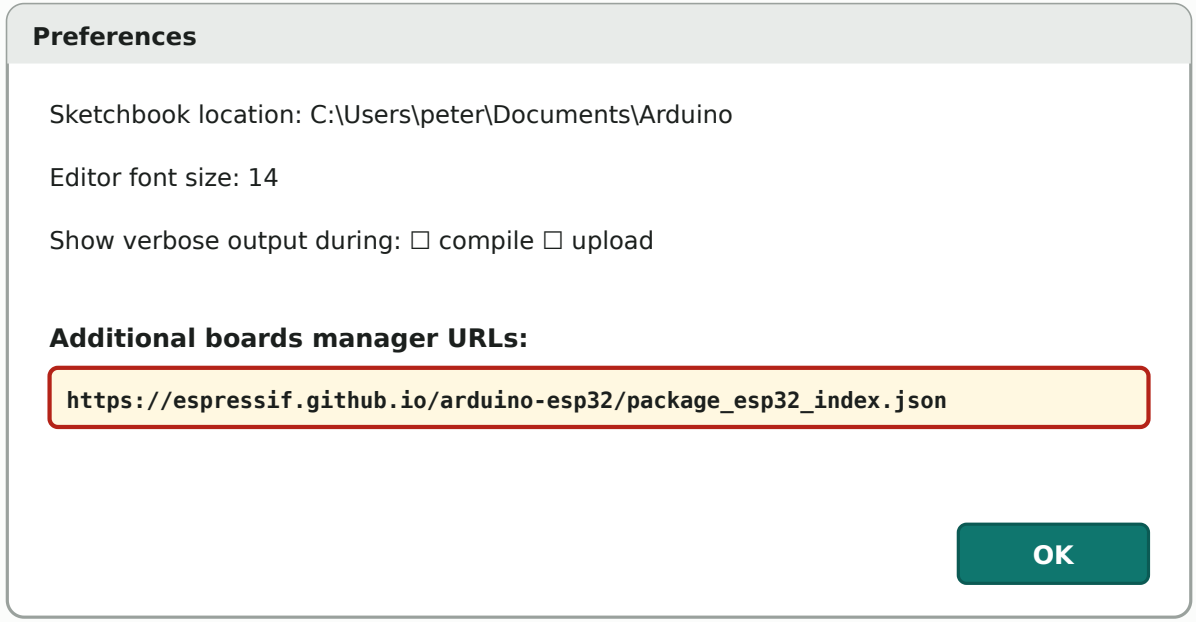


Figure 21. The ESP32 board (7 × 9 cm). Two rails (GND and 3V3), two pull-up resistors, and solder points for the sensor and modem wires.

STAGE 18: INSTALL THE ARDUINO IDE AND DRIVERS



Where: File → Preferences
(Mac: Arduino IDE → Settings)
Paste the address into the yellow box.
If something is already there, put a comma after it and paste yours after the comma.
Click OK.

Figure 22. Adding the ESP32 boards address in Preferences. You do this once per computer.

STAGE 18: INSTALL THE ARDUINO IDE AND DRIVERS

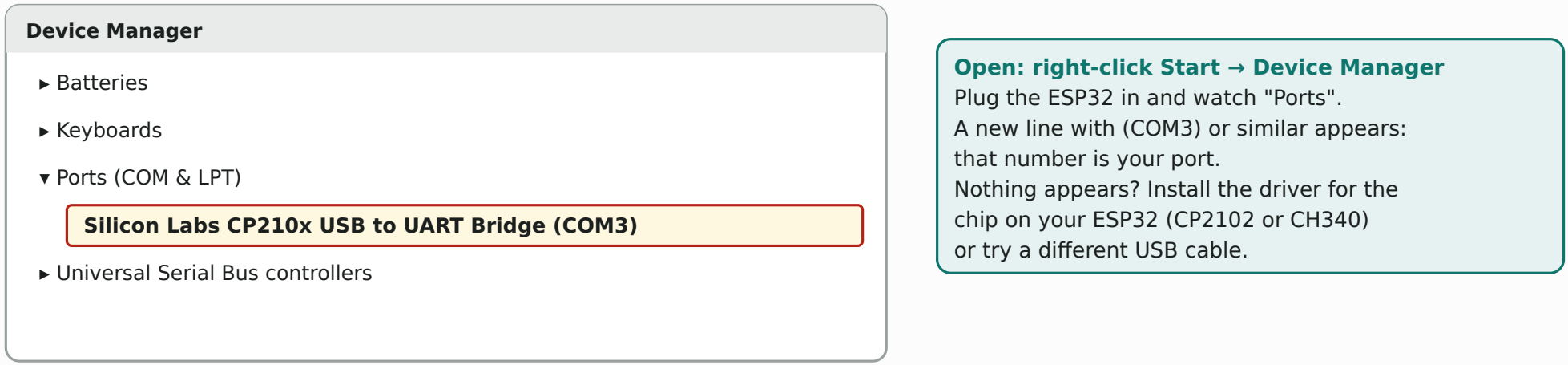


Figure 23. Finding the ESP32's COM port on Windows. The name says CP210x or CH340 depending on the USB chip on your board.

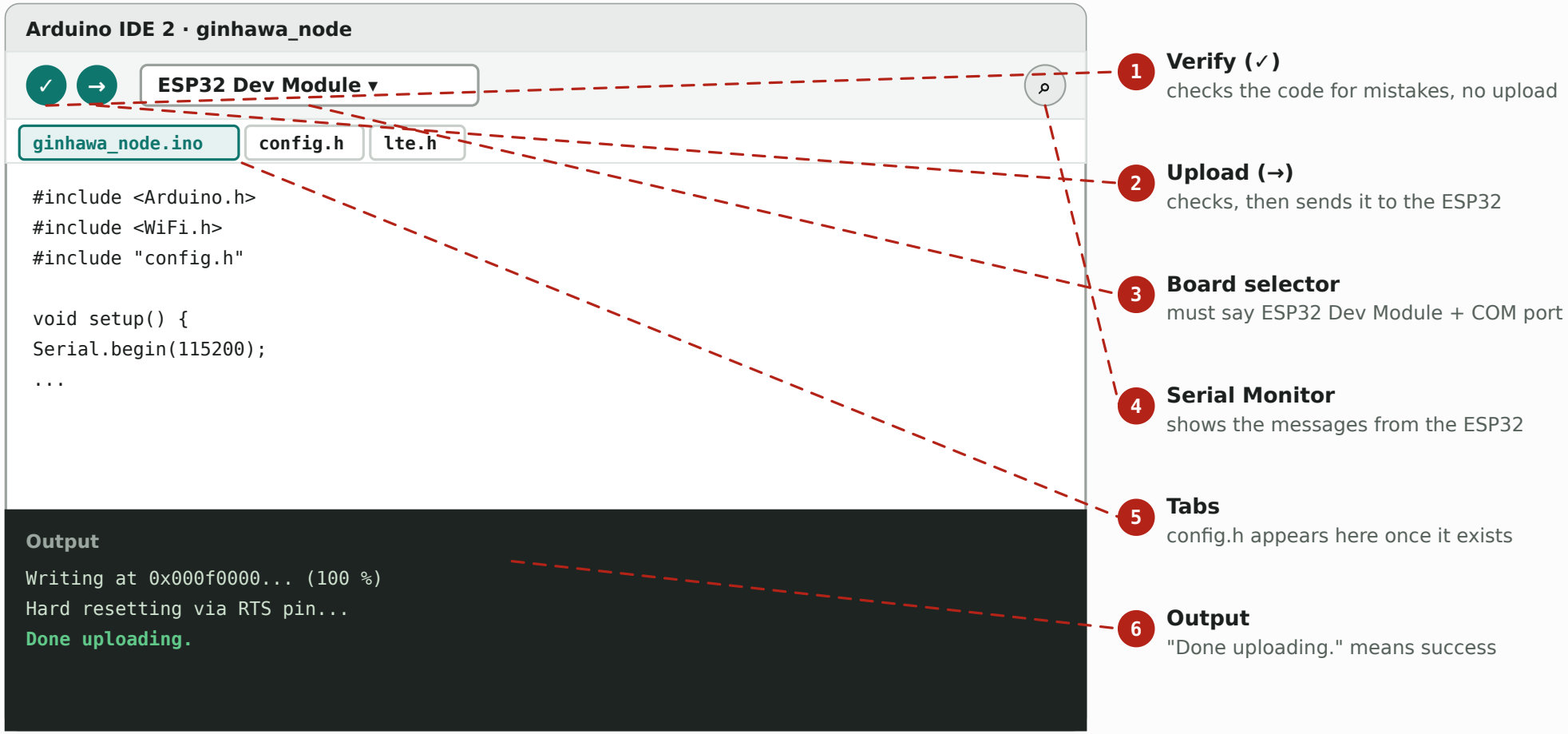


Figure 24. The Arduino IDE 2 window and the six things you will use. (Simplified drawing; colours on your screen may differ.)

STAGE 20: CONNECT THE SEN55 AND INA219, THEN TEST

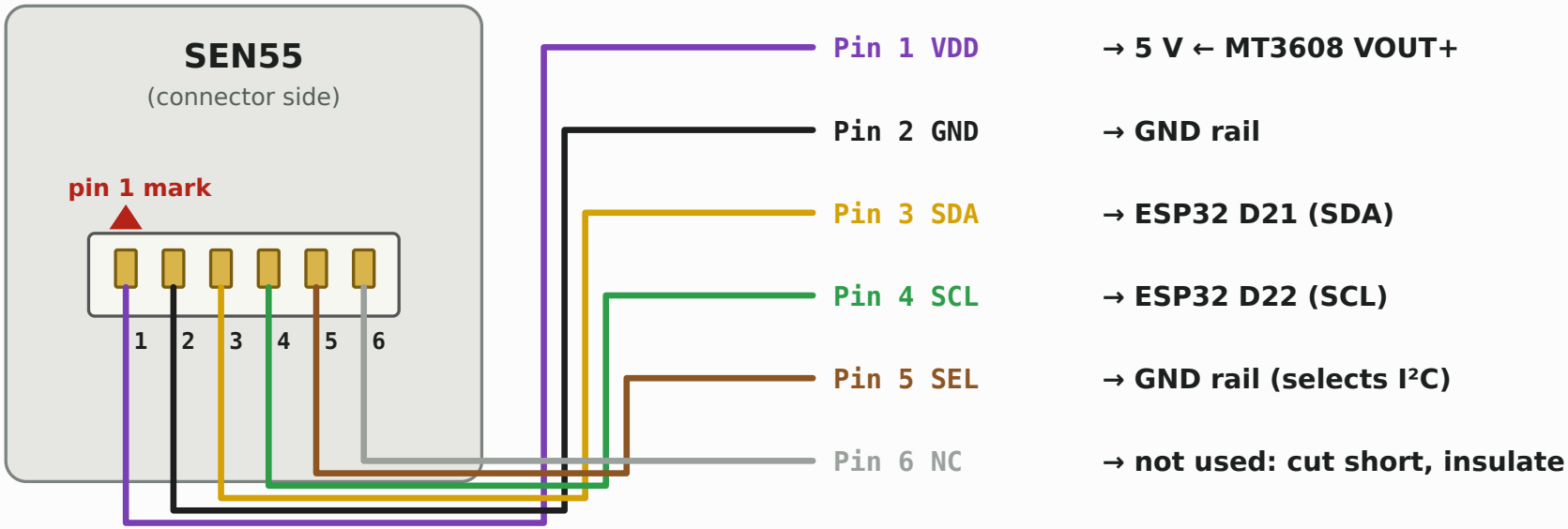


Figure 26. SEN55 pin-out and where each wire goes. Find **pin 1** from the mark printed next to the sensor's connector, then count. Go by pin number, never by wire colour.

STAGE 21: CONNECT THE 4G MODEM (A7670SA) AND TEST

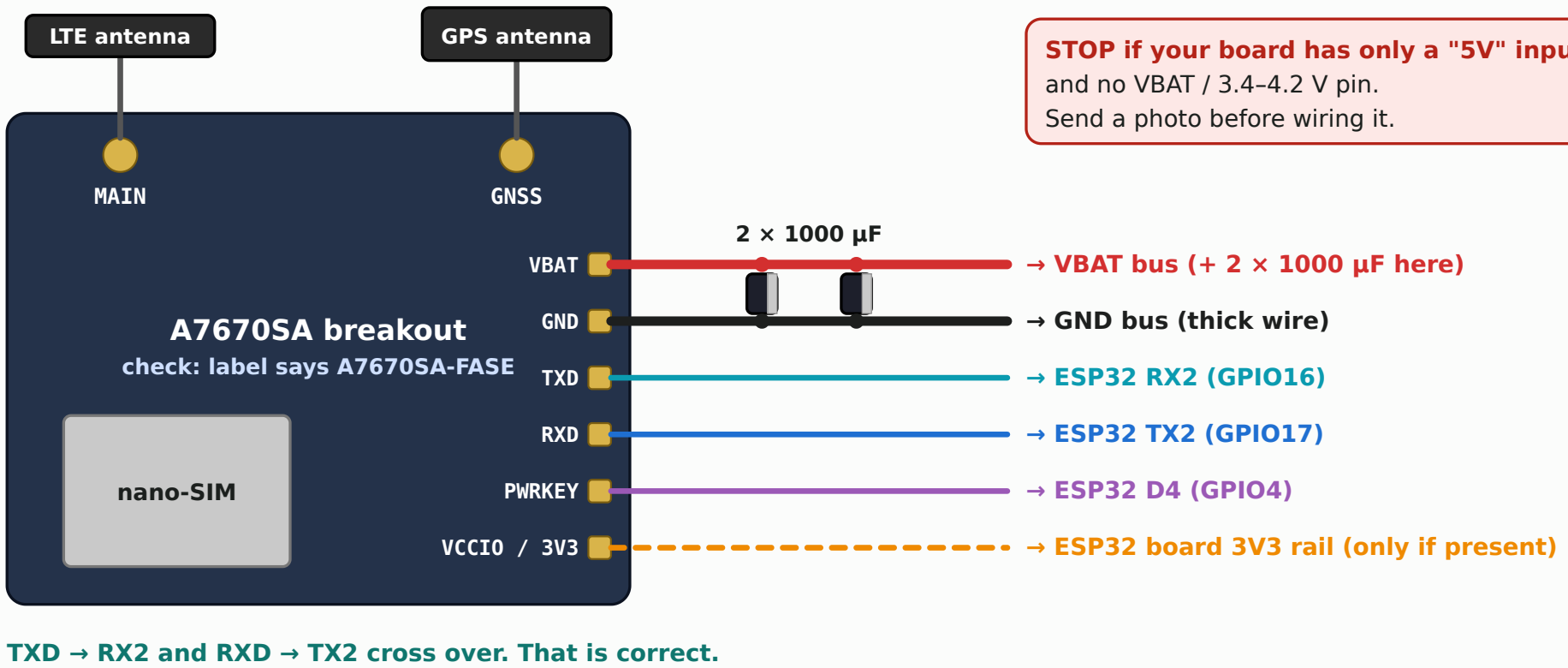


Figure 27. A7670SA connections. Pin names differ between breakout makers: PWRKEY may be printed as **PWR**, **PEN** or **K**; VBAT may be **VCC 3.4–4.2 V** or **BAT**.

STAGE 21: CONNECT THE 4G MODEM (A7670SA) AND TEST

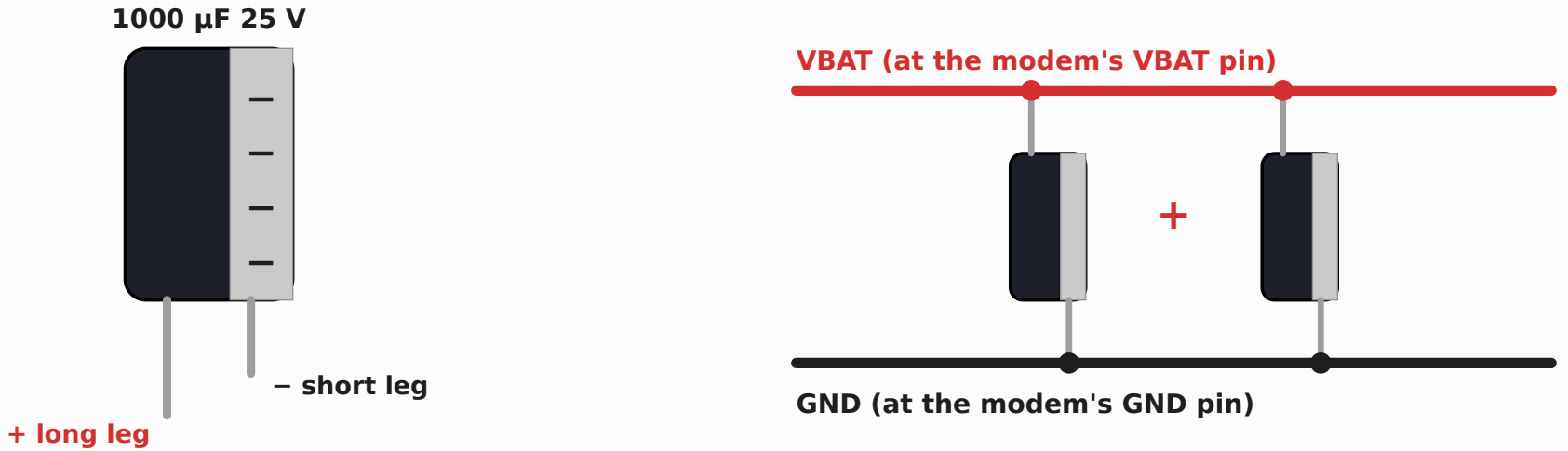


Figure 28. Capacitor polarity. The side with the stripe is **−**. Two capacitors side by side, both + to VBAT and both − to GND.

STAGE 23: UPLOAD THE GINHAWA FIRMWARE

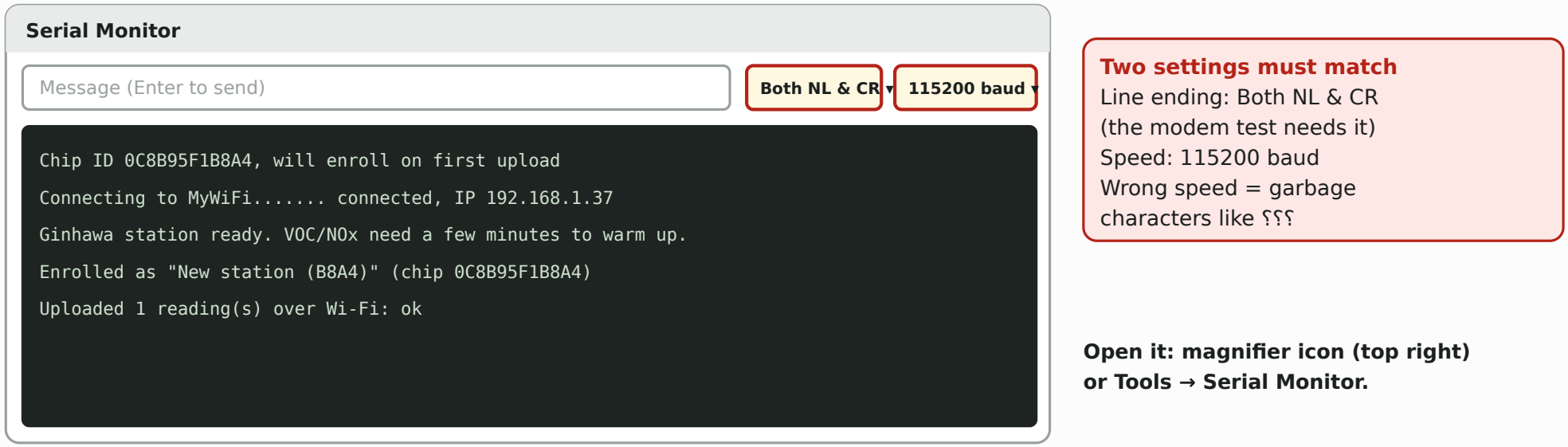
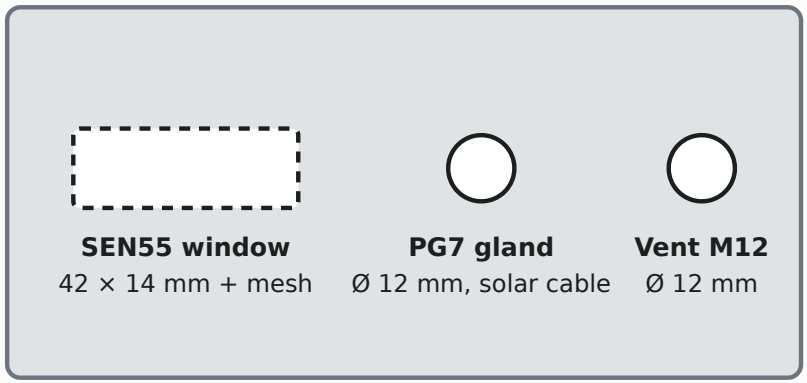


Figure 25. The Serial Monitor shows what the ESP32 is doing. The text is an example of a healthy first start.

STAGE 25: PREPARE THE ENCLOSURE

Bottom face (150 × 70 mm box shown)



Drill on the bottom (or low on the sides).
Never drill the top: rain sits there.

ABS / plastic boxes let radio signals through.
Flat or patch antennas can stay inside the box.
Only the solar cable needs a gland.

Inside, lid off, box standing upright

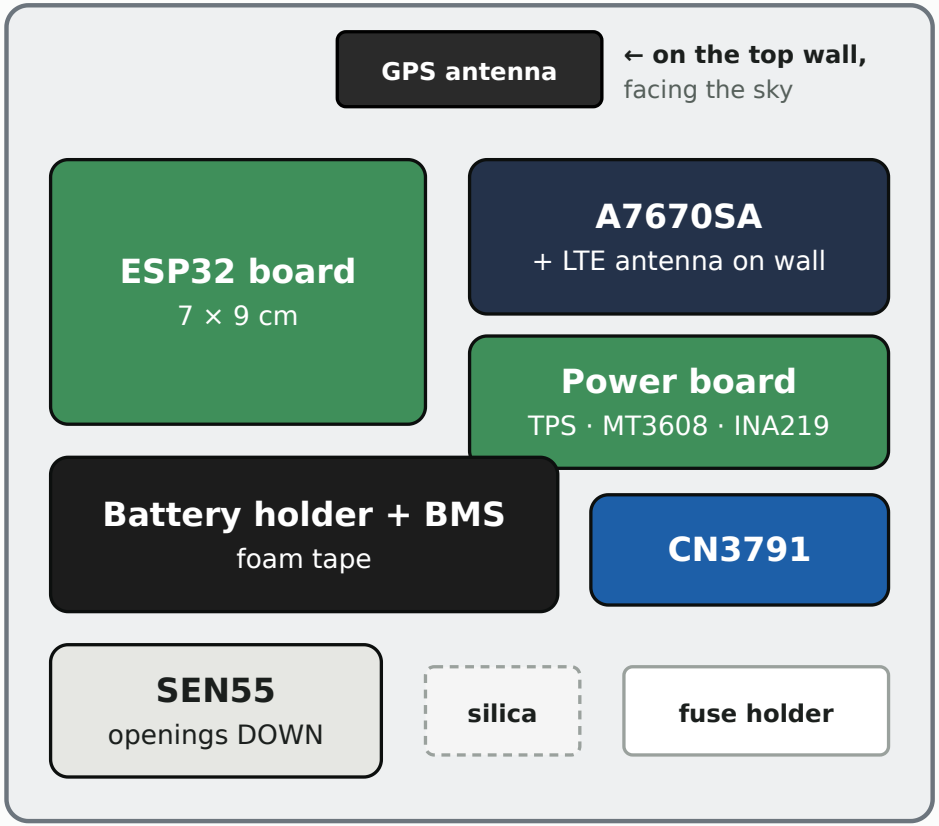


Figure 30. Drilling plan and layout inside an IP65 box of about 150 × 100 × 70 mm. Adjust the positions to the box you buy, keeping these rules.

STAGE 25: PREPARE THE ENCLOSURE

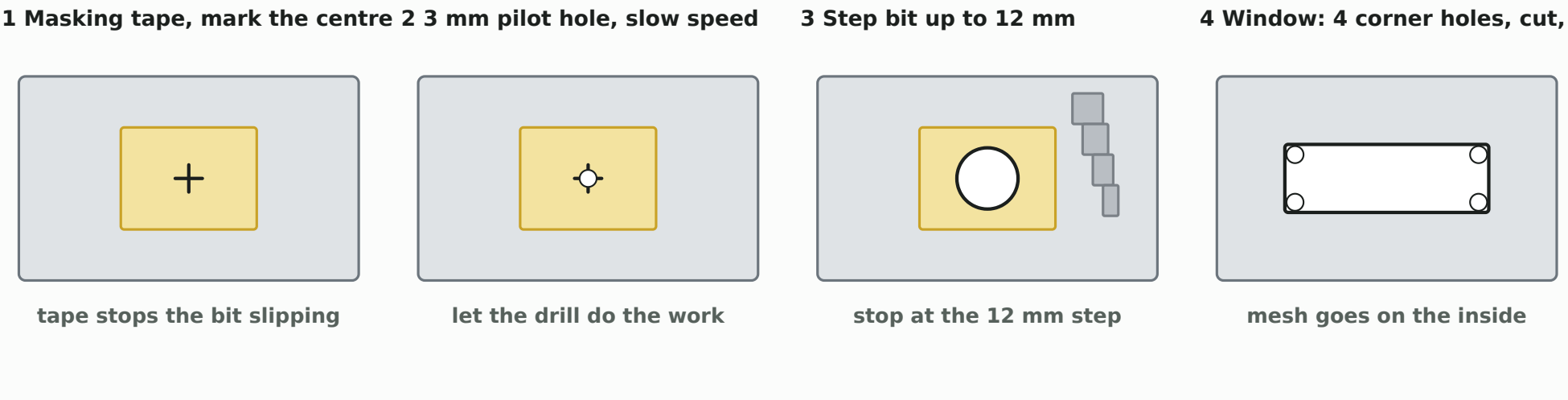


Figure 29. Drilling the plastic box. Go slowly: plastic cracks if you push hard or drill fast.

STAGE 27: MOUNT IT OUTSIDE

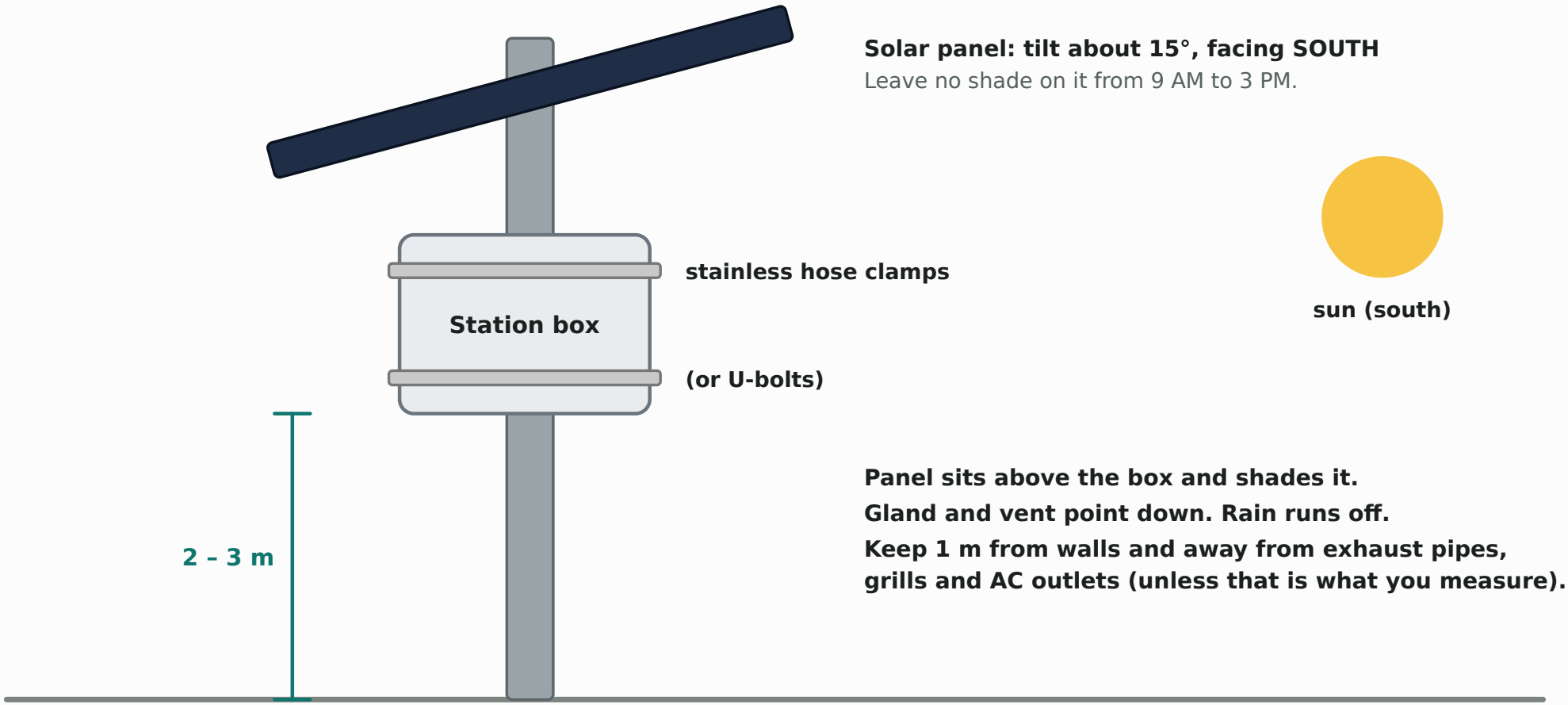


Figure 31. Mounting on a pole. The sensor should breathe the same air people do: 2 to 3 m above the ground, in open air.